MICROCONTROLLER INTERFACING

Digital-to-Analog Converter (DAC)

of STM32F446RE Nucleo

GROUP NO. 13

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12-BIT • TWO CHANNELS • DMA

From digital code to analog voltage

- 12-bit voltage-output converter; configurable for 8- or 12-bit operation
- 12-bit data supports left or right alignment
- Two independent output channels and simultaneous dual conversion
- VREF+ defines the positive reference; DMA support is available





A discrete input code is translated into a proportional output level.

DAC main features

01 12-bit output
Two channels

02 8 / 12-bit
Left / right aligned

03 Synchronized
update

04 Noise-wave
generation

05 Triangle-wave
generation

06 Independent or simultaneous
dual conversion

07 DMA per channel
Underrun detection

08 External trigger
selection

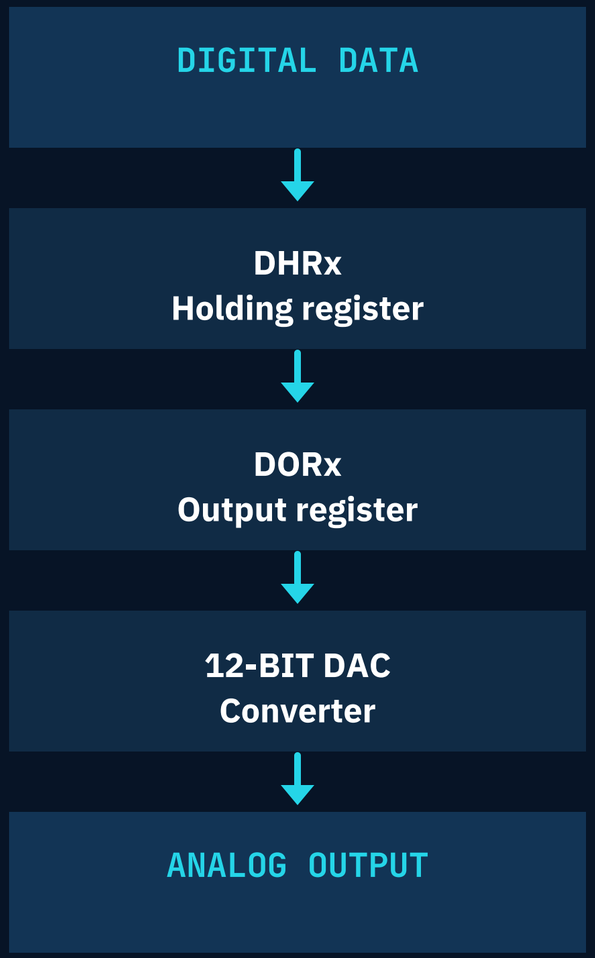
09 VREF+ voltage
reference

Two converters, two buffered outputs



- Each channel owns a converter
- Buffer lowers output impedance and may be disabled
- PA4 / PA5 expose DAC_OUT1 / DAC_OUT2 when enabled

Registers stage every conversion



Software writes conversion data to DAC_DHRx.

DAC_DORx is read-only; the selected DHR value is copied here.

Analog output becomes valid after converter settling time.

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NO TRIGGER  ·  DHRx → DORx after 1 APB1 cycle
HARDWARE TRIGGER  ·  Trigger → update after 3 APB1 cycles
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Choose precision and alignment

8-bit Right Alignment

DAC_DHR8Rx

DATA BITS [7:0]

12-bit Left Alignment

DAC_DHR12Lx

DATA BITS [15:4]

12-bit Right Alignment

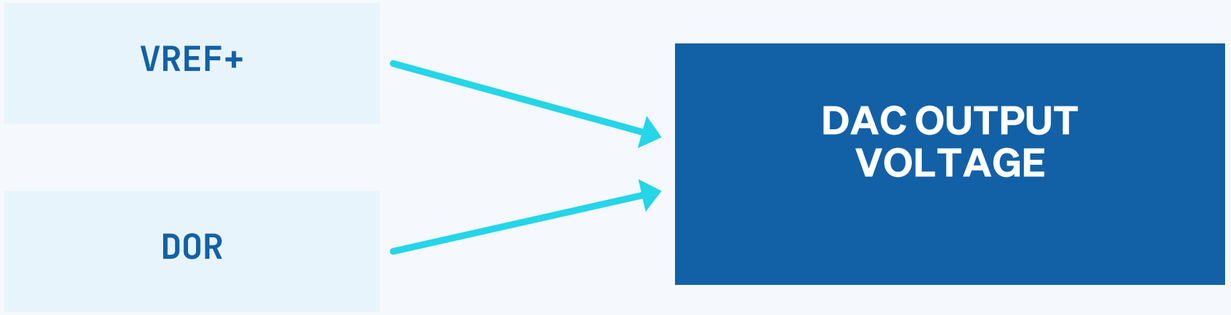
DAC_DHR12Rx

DATA BITS [11:0]



A linear relationship to VREF+

$$\text{DAC output} = (\text{VREF+} \times \text{DOR}) / 4096$$



Digital input codes map linearly to voltages from 0 to VREF+. The final output depends directly on the DAC output register value.

Triggers define exactly when output updates

Trigger Enable
TENx

↓

Trigger Selection
TSELx

↓

Trigger Event

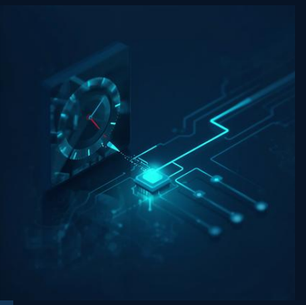
↓

DHRx → DORx

↓

DAC Output

Timer 6 TRGO	Timer 8 TRGO
Timer 7 TRGO	Timer 5 TRGO
Timer 2 TRGO	Timer 4 TRGO
EXTI Line 9	Software Trigger



TSELx[2:0] selects the trigger source

Automated transfers—with underrun protection



- Each DAC channel supports DMA
- DMAENx enables requests on external triggers
- Requests are not queued; a new trigger during service causes underrun
- DMAUDRx reports the error; conversion continues with old data

TRIGGER RATE TOO HIGH → DMA MISSES DEADLINE → UNDERRUN → DMAUDRx SET

Wave generation and dual-channel control

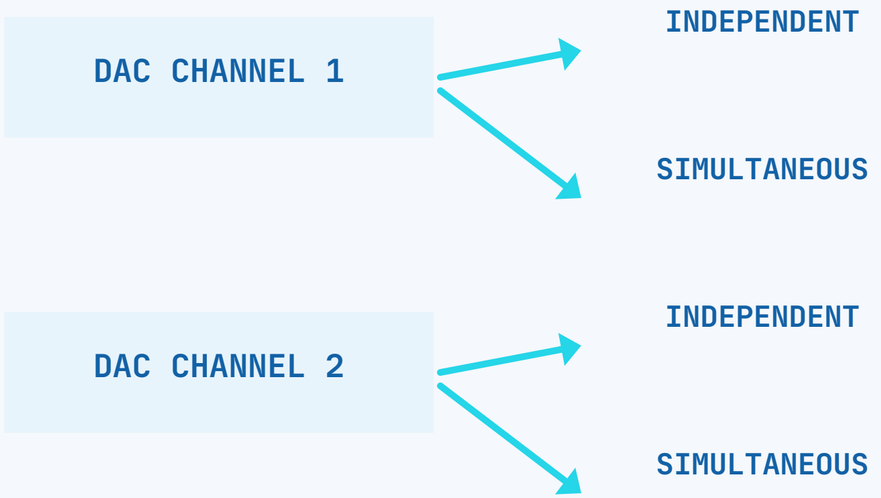
WAVE GENERATION

NOISE GENERATION
WAVE = “01” · LFSR
Trigger must be enabled

TRIANGLE GENERATION
WAVE = “10” · amplitude via MAMP
Triggered internal triangle counter

`DAC_DHRx + WAVE VALUE → DAC_DORx`

DUAL DAC CONVERSION



DUAL REGISTERS · DHR8RD / DHR12RD / DHR12LD
Single access updates both channels

CONCLUSION

One flexible analog-output subsystem



The STM32F446RE DAC provides flexible digital-to-analog conversion through two channels, configurable data formats, trigger control, DMA capability, and wave-generation modes.

THANK YOU

Questions?